

Covenant Screening Under IFRS-16: Assumption-Bounded Analytic Approximations and Simulation-Based Evaluation

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Revised manuscript

Abstract

We compare IFRS-16-inclusive and stipulated frozen-GAAP covenant calculations using a reduced-form analytic screening model and a full annual financing simulation. The latter funds opening cash through closing sources and uses and reconciles mandatory principal, revolver drawings, cash sweeps and payment default. Deterministic screening statements apply only to matched reduced-form paths with established error budgets and positive denominator floors; empirical cross-model discrepancies also reflect structural, timing and input differences. In the current seed-42 synthetic benchmark of 200 five-year scenarios, the analytic ranking score has AUC 0.9472 (scenario-bootstrap 95% interval [0.8974, 0.9828]), but the 0.5 classification threshold has an 85% false-negative rate and a zero false-positive rate. Twenty scenarios financially fail, including 17 payment defaults. Mean scenario RMSE is 0.8084 for leverage and 2.3157 for coverage; minimum-headroom RMSE is 1.4884. A separate Accor accounting-convention illustration reproduces supplied reported/reconstructed inputs and hypothetical thresholds, without claiming predictive validation. The score is not a calibrated default probability, and the benchmark supplies no equity-return or optimization uplift result. Code, input assumptions, numerical artifacts and six regenerated figures accompany the revised manuscript.

Keywords: IFRS-16, leveraged buyout, covenant screening, synthetic benchmark

JEL Classification: G24, G32, C61

1 Introduction

Lease capitalization changes the quantities entering debt covenants, while a credit agreement’s definitions determine which accounting effects enter its tests. IFRS 16, issued in 2016 and effective for annual periods beginning in 2019, brings most lessee lease commitments onto the balance sheet, subject to its exemptions ([IFRS Foundation, 2016](#)). Comparing a lease-inclusive covenant calculation with a stipulated rent-adjusted alternative is therefore useful for lease-intensive transactions. Neither calculation by itself establishes the terms of an actual loan.

This paper studies assumption-bounded analytic screening alongside an explicit annual financing simulation. These are two materially different model families. The analytic model provides transparent debt and coverage paths. The simulation reconciles closing funding, operating cash, debt service, revolver capacity and cash retention. Differences between their outputs combine approximation error, structural-model differences, timing differences and input/convention differences. An empirical comparison cannot isolate the first component without a separate, matched-model experiment.

1.1 Contributions and scope

The contribution is a reproducible characterization of this screening workflow, conditional mathematical statements about matched reduced-form paths, and an evaluation on synthetic scenarios. We distinguish a ranking score from an operational decision rule, and distinguish covenant compliance from the broader financial-failure union. An Accor SA accounting-convention illustration retains the link to a named company while making reconstructed inputs and hypothetical thresholds explicit. The present evaluation does not fit a borrower prediction model or estimate the return from an optimized covenant package.

The revision replaces historical empirical claims after financing corrections. In particular, retained opening cash is a funded use, so sponsor equity and any return calculated from it must use the repaired closing bridge. Historical IRR tables cannot be carried forward. The title accordingly emphasizes screening and simulation-based evaluation rather than an empirically demonstrated optimization or a universal deterministic guarantee.

2 Related Work and Background

2.1 Covenants and control rights

Dichev and Skinner (2002) investigate the debt covenant hypothesis using a large sample of private lending agreements. Nini et al. (2009) study creditor control rights and investment restrictions outside payment default. These studies motivate careful separation of covenant conditions from missed debt service; they do not validate the synthetic data-generating process used here. The present exercise concerns model behavior under stipulated inputs, rather than estimating contracting effects in those borrower populations.

2.2 Lease reporting and covenant conventions

IFRS 16 specifies lease reporting, whereas contractual covenant definitions may adjust accounting quantities. We use the repository’s two named conventions as explicit formulas. Its frozen-GAAP helper adds supplied rent to supplied EBITDA for coverage and excludes leases from net debt. This is a stipulated rent-adjusted coverage convention, not an automatic reconstruction of historical-GAAP accounts. Historical operating-lease EBITDA deducts operating rent; IFRS-16 EBITDA generally does not include that same operating rent charge. Inputs must therefore be aligned with the intended credit agreement before using either helper.

3 Notation and Constants

Table 1: Notation and units. Superscript a identifies the analytic family.

Symbol	Unit	Definition
t, T	years	Annual index and horizon; empirical $T = 5$.
B_t, R_t, L_t, C_t	currency	Closing term debt, revolver, lease liability and cash.
E_t, V_t	currency/year	EBITDA and revenue.
I_t^d, I_t^L	currency/year	Financial interest and lease interest.
A_t, U_t, S_t	currency/year	Actual mandatory amortisation, unpaid amount and sweep.
X_t^a, X_t^ℓ, Q_t	currency/year	Amortisation draw, liquidity draw and revolver repayment.
s, q, η	fractions	Sweep, lease principal and lease-addition coefficients.
$c^{\text{lev}}, c^{\text{icr}}$	ratios	Maximum leverage and minimum coverage.
h_t, H	ratio points	Minimum of two covenant margins; minimum over time.
ϵ_Z	units of Z	Assumed or established error budget for component Z .

Combining leverage and coverage margins by their numerical minimum is the implemented screening convention. The two margins have different economic interpretations despite both being dimensionless ratios; this minimum is not a monetary distance to default.

4 Model Framework

4.1 Dual covenant conventions

For positive EBITDA and positive applicable interest denominators, the project calculates

$$\text{ND}_t^{16} = B_t + R_t + L_t - C_t, \quad \text{Lev}_t^{16} = \frac{\text{ND}_t^{16}}{E_t}, \quad \text{ICR}_t^{16} = \frac{E_t}{I_t^d + I_t^L}, \quad (1)$$

$$\text{ND}_t^F = B_t + R_t - C_t, \quad \text{Lev}_t^F = \frac{\text{ND}_t^F}{E_t}, \quad \text{ICR}_t^F = \frac{E_t + P_t}{I_t^d}, \quad (2)$$

where P_t is supplied rent in the frozen-GAAP helper. The full synthetic benchmark evaluates only the IFRS-16-style view in (1); the separate Accor illustration evaluates both. Nonpositive EBITDA is a failure condition rather than evidence of favorable negative leverage. The Accor helpers return undefined ratios in that state. The benchmark instead checks the failure flag separately and floors the interest denominator at 10^{-6} in calculating simulated coverage.

4.2 Reduced-form analytic screening

The implemented paths use $E_t^a = E_0^a(1+g)^t$ and $F_t^a = (\alpha - \kappa)E_t^a$, with weighted financial rate $r_d = wr_{\text{sen}} + (1-w)r_{\text{mezz}}$. They obey

$$D_{t+1}^a = \max\{0, (1+r_d)D_t^a - sF_t^a\}, \quad (3)$$

$$L_{t+1}^a = \max\{0, (1-q)L_t^a + \eta E_t^a\}, \quad (4)$$

$$C_0^a = C_{\text{initial}}^a, \quad C_t^a = C_{\text{min}}^a \quad (t \geq 1). \quad (5)$$

Equation (4) is the run-off mode used by the benchmark. The alternative steady-state mode sets $L_t^a = \lambda E_t^a$. Where the debt floor does not bind, the familiar linear-recurrence expression is

$$D_t^a = D_0^a(1+r_d)^t - s \sum_{k=0}^{t-1} (1+r_d)^{t-1-k} F_k^a. \quad (6)$$

After a floor binds, the implemented recursion, rather than an unrestricted closed-form extrapolation, defines the path. Analytic coverage uses closing debt and lease balances. Interest capitalization in (3) and fixed cash are reduced-form assumptions; there is no reconciled mandatory-amortisation or revolver waterfall in this family.

4.3 Full simulation: closing and operating cash

For stipulated debt-free purchase use V_{purchase} , fee fraction f and retained opening cash C_0 , sponsor funding is

$$E_{\text{sponsor}} = V_{\text{purchase}}(1+f) + C_0 - B_0. \quad (7)$$

Closing debt plus sponsor equity equals total uses. Opening cash is not a source of acquisition funding. Debt above uses is rejected; zero sponsor equity remains exactly zero, with undefined IRR and MOIC. Default assumptions provide a mechanical check: purchase 1,000, fees 30 and cash 40 are funded by debt 450 and equity 620. These are model currency units, not observed transaction amounts.

Revenue in year 1 equals the supplied starting revenue; growth begins in year 2: $V_t = V_0(1+g)^{t-1}$ and $E_t = mV_t$. Using opening balances,

$$I_t^d = r_d^s(B_{t-1} + R_{t-1}), \quad I_t^L = r_L^s L_{t-1}, \quad (8)$$

$$\text{Tax}_t = \max\{0, \tau(E_t - dV_t - I_t^d - I_t^L)\}, \quad (9)$$

$$O_t = E_t - \Delta WC_t - \text{Tax}_t - I_t^d - I_t^L - kV_t, \quad (10)$$

$$K_t = C_{t-1} + O_t - qL_{t-1}, \quad L_t = \max\{0, (1-q)L_{t-1} + \eta V_t\}. \quad (11)$$

Here K_t is cash before financing. Working capital is a fixed revenue fraction; the initial working-capital balance makes year-1 ΔWC zero. Lease additions are noncash and proportional to current revenue. This stylized schedule is not a lease-by-lease IFRS valuation or a complete financial-statement model.

4.4 Full simulation: financing waterfall

Let $M_t = \min\{M, B_{t-1}\}$ denote scheduled principal and $U_t^R = \max\{0, R_{\max} - R_{t-1}\}$ available revolver capacity. The waterfall is

$$A_t^{\text{cash}} = \min\{M_t, \max(0, K_t)\}, \quad (12)$$

$$X_t^a = \min\{M_t - A_t^{\text{cash}}, U_t^R\}, \quad A_t = A_t^{\text{cash}} + X_t^a, \quad (13)$$

$$U_t = M_t - A_t, \quad K'_t = K_t - A_t^{\text{cash}}, \quad (14)$$

$$X_t^\ell = \min\{\max(0, C_{\min} - K'_t), \max(0, U_t^R - X_t^a)\}. \quad (15)$$

Thus amortisation drawings go directly to the term lender. They do not increase retained cash. A negative operating balance remains negative unless a separate liquidity draw funds it. After these draws, excess cash repays the revolver first. If $K_t'' = K'_t + X_t^\ell - Q_t$ is cash after that repayment, the optional sweep is

$$S_t = \min\{\max(0, K_t'' - C_{\min}), s \max(0, K_t), \max(0, B_{t-1} - A_t)\}. \quad (16)$$

The resulting accounting identities are

$$C_t = C_{t-1} + O_t - qL_{t-1} - A_t - S_t + X_t^a + X_t^\ell - Q_t, \quad (17)$$

$$B_t + R_t = B_{t-1} + R_{t-1} - A_t - S_t + X_t^a + X_t^\ell - Q_t. \quad (18)$$

Unpaid mandatory principal stays outstanding. Payment default means $U_t > 0$. Reserve failure means $C_t < C_{\min}$, not a legal insolvency finding. Mandatory principal takes priority over the reserve. Simulation continues after failure for diagnostics, without a recovery or arrears catch-up model.

4.5 Covenant design and an unevaluated return objective

The historical conceptual objective can be written, for a decision vector x and explicit uncertainty distribution $\mathcal{P}$, as

$$\begin{aligned} & \sup_x \quad \mathbb{E}_{\theta \sim \mathcal{P}}[\text{IRR}(x; \theta)] \\ & \text{subject to} \quad \text{Lev}_t^a(x) + \epsilon_{\text{lev}}(t) \leq c^{\text{lev}}, \\ & \quad \quad \quad \text{ICR}_t^a(x) - \epsilon_{\text{icr}}(t) \geq c^{\text{icr}}, \quad t = 1, \dots, T. \end{aligned} \quad (19)$$

This is a possible reduced-form design formulation, not a solved empirical optimization in the current benchmark. If only thresholds change and financing and cash flows are fixed, an individual

deal's return does not change. Claims of return improvement require a defined financing decision and reoptimized feasible set. The implementation's sponsor vector is $[-E_{\text{sponsor}}, 0, \dots, 0, E_{\text{exit}}]$, with exit EBITDA multiple less sale costs, financial debt and leases, plus cash. No equity-return output is emitted by the benchmark, so no expected-IRR or IRR-improvement estimate is reported here.

4.6 Bayesian hierarchical calibration: separate research framework

The earlier framework contemplated bounded-support hierarchical parameters. For bounds $a_j < b_j$, one possible specification is

$$z_{ij} \sim \mathcal{N}(\mu_j, \sigma_j^2), \quad \theta_{ij} = a_j + (b_j - a_j)\{1 + \exp(-z_{ij})\}^{-1}, \quad (20)$$

$$\log \Lambda_i \sim \mathcal{N}(\mu_\Lambda, \sigma_\Lambda^2), \quad \sigma_j \sim \text{HalfNormal}(s_j). \quad (21)$$

Growth shocks and cross-parameter dependence would require additional specified distributions. These equations preserve the conceptual distinction between parameter uncertainty and financial dynamics. They are not a description of a posterior estimated for the present results. The separate calibration module has optional PyMC and MAP/Laplace paths; the benchmark calls neither. It samples clipped scenario parameters directly. Consequently this paper reports no posterior credible frontier or empirical benefit from Bayesian calibration.

5 Deterministic Approximation Bounds

5.1 Scope of the guarantees

We retain deterministic reasoning only for a pair of matched reduced-form paths with established component error budgets and positive denominators. A reference path below is not automatically the full financing simulation. Assumptions in Appendix B are requirements to be checked, not consequences of the current synthetic sampler. In particular, no universal financial-default guarantee follows from bounding two covenant ratios.

Proposition 1 (Conditional ratio screening). *Suppose reference and analytic EBITDA and interest satisfy $E, E^a \geq e > 0$ and $I, I^a \geq i > 0$ at a given date. Suppose $|E^a - E| \leq \epsilon_E$, $|N^a - N| \leq \epsilon_N$, $|I^a - I| \leq \epsilon_I$, and known upper bounds are $|E^a| \leq \bar{E}$ and $|N^a| \leq \bar{N}$. Then*

$$|\text{ICR}^a - \text{ICR}| \leq \epsilon_{\text{icr}} := \frac{\epsilon_E}{i} + \frac{\bar{E}\epsilon_I}{i^2}, \quad (22)$$

$$|\text{Lev}^a - \text{Lev}| \leq \epsilon_{\text{lev}} := \frac{\epsilon_N}{e} + \frac{\bar{N}\epsilon_E}{e^2}. \quad (23)$$

For matched debt and lease definitions and fixed rates, $\epsilon_N = \epsilon_D + \epsilon_L + \epsilon_C$ and $\epsilon_I = r_d\epsilon_D + r_L\epsilon_L$ suffice.

Cash cannot be omitted from net-debt error unless it is identical between paths. Different interest timing or rates require their own additional error budgets. For the rent-adjusted coverage numerator, a rent error must likewise enter the numerator budget. The proposition is deliberately conditional; the constants emitted by `analytic_bounds.py` do not establish its premises.

Proposition 2 (Feasible-set monotonicity). *Fix the cash-flow mapping, testing dates, coverage threshold and objective. If $c_1^{\text{lev}} \leq c_2^{\text{lev}}$, the feasible sets defined by the same screening inequalities satisfy $\mathcal{F}(c_1^{\text{lev}}) \subseteq \mathcal{F}(c_2^{\text{lev}})$. Their supremum objective values are therefore non-decreasing, when defined.*

This is set inclusion, not a claim that relaxing a covenant changes a fixed transaction’s IRR or induces a specific borrower response.

Proposition 3 (Conditional covenant certification). *Under Proposition 1 at every testing date, let $h_t^a = \min\{\text{ICR}_t^a - c^{\text{icr}}, c^{\text{lev}} - \text{Lev}_t^a\}$ and $\epsilon_{\max} = \max_t \max\{\epsilon_{\text{icr}}(t), \epsilon_{\text{lev}}(t)\}$. If $h_t^a > \epsilon_{\max}$ at every date, the reference path strictly satisfies both ratio covenants at those dates.*

This statement does not certify reserve adequacy, payment of principal, or the outcomes of the unmatched full simulation. No empirical certification percentage is reported.

6 IFRS-16 LBO Benchmark Generator

6.1 Synthetic population and executable protocol

The authoritative procedure is `analysis/run_benchmark.py` in the repository (Bhardwaj, 2026). With seed 42, it samples 200 scenarios from five synthetic operator templates with replacement, each with five annual observations. The 20-scenario smoke mode is only a separate software check. The metadata file `scenario_parameters.csv` is checksummed but does not control sampling. There is no audited borrower or transaction panel.

For each scenario, the sampler draws a base, downside or distressed regime with probabilities (0.5, 0.3, 0.2). Growth is initially normal around the operator’s mean with its supplied standard deviation and clipped to $[-0.12, 0.18]$; margin has standard deviation 0.02 and is clipped to $[0.10, 0.35]$. Sweep has standard deviation 0.04 and range $[0.30, 0.85]$. Lease principal and additions have standard deviations 0.01 and 0.003 and ranges $[0.05, 0.20]$ and $[0, 0.05]$ respectively. Downside then subtracts 0.04 from growth and 0.03 from margin, multiplies debt by 1.05, leases by 1.04 and cash by 0.85. Distress uses respective changes $-0.08, -0.06, 1.20, 1.10, 0.60$. Final growth and margin floors are -0.12 and 0.08 . Draws are constant scenario parameters, not five new annual random shocks.

Table 2: Material distinctions in the actual benchmark model pair.

Feature	Analytic screening	Full simulation
Initial EBITDA	Operator template EBITDA	Revenue times drawn margin
Growth timing	Applied at time 1	Starts in year 2
Cash	Initial 25; fixed 10 thereafter	Regime-adjusted template cash; reserve 25
Financial interest	Weighted 0.062; closing debt	0.06; opening term debt and revolver
Lease interest	0.045; closing leases	0.05; opening leases
Lease additions	Prior EBITDA basis	Current revenue basis
Debt service	Capitalization and FCF sweep	Cash interest; mandatory principal 30; revolver
Revolver capacity	None	100 distressed; 200 otherwise
Operating deductions	$(\alpha - \kappa)E^a$	Explicit tax, working capital and capex

Analytic α is 0.72 in base and 0.68 otherwise; κ is 0.10 in distress and 0.08 otherwise. Full-simulation depreciation, tax, capex and working-capital fractions are 0.03, 0.25, 0.04 and 0.02. Full scenario assumption records are archived alongside paths, so the table is not a substitute for executable inputs.

6.2 Failure outcome, score and uncertainty

The outcome is the union, over all five years, of nonpositive EBITDA, payment default, reserve funding deficit, leverage above 6.0 or coverage below 1.8. Equality at a covenant threshold is

not a simulated breach. The minimum analytic headroom and ranking score are

$$H^a = \min\{6 - \max_t \text{Lev}_t^a, \min_t \text{ICR}_t^a - 1.8\}, \quad z = (1 + \exp(2H^a))^{-1}. \quad (24)$$

The score is not a calibrated probability of default. Classification declares failure for $z \geq 0.5$, including exactly zero analytic headroom. Thus the decision boundary’s equality convention differs from the simulation’s strict breach test. No model fitting, train/test split, cross-validation or threshold optimization is used in this evaluation.

AUC is computed across scenarios. Its percentile 95% interval uses 500 scenario bootstrap attempts with seed 42, skipping single-class resamples. This is not an operator-clustered interval. Repeated scenarios share five template assumptions, so the interval does not establish out-of-sample borrower performance. There are no RMSE confidence intervals, permutation tests, calibration curves or Brier scores in the current report. Execution exceptions and a single-class full sample abort reporting instead of silently removing scenarios.

7 Empirical Validation

7.1 Analytic versus simulation diagnostics

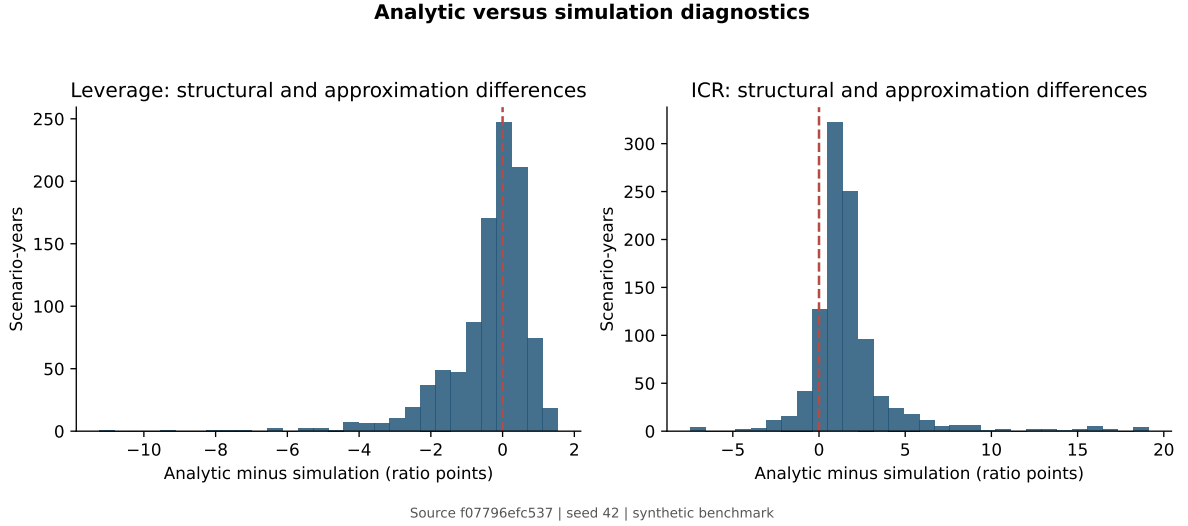


Figure 1: Replacement for the historical theoretical-guarantees plot: signed analytic-minus-simulation leverage and ICR differences over 1,000 scenario-years. Zero is an equality reference, not an error bound. The distributions combine structural, timing, input and approximation differences; they do not test a universal deterministic guarantee. Source: archived regenerated scenario paths, seed 42; exact source revision in Appendix E.

Table 3: Current cross-model discrepancy metrics, in ratio points.

Metric	Value
Leverage MAE	0.784173
Mean scenario leverage RMSE	0.808420
ICR MAE	2.011110
Mean scenario ICR RMSE	2.315683
Minimum-headroom MAE	0.777709
Minimum-headroom RMSE	1.488412

Leverage and ICR MAE are averaged across scenarios after computing each scenario’s five-year MAE. Their reported RMSE is the mean of the 200 scenario-level five-year RMSEs, not the pooled root mean square across all 1,000 observations. Minimum headroom is one scalar per scenario; its RMSE is the square root of the mean of 200 squared headroom differences. This distinction is necessary for exact replication. The old headroom table used a different and unreproduced protocol; Table 3 is a new evaluation, not an update to that table’s cells.

7.2 Benchmark performance analysis

The AUC is 0.947222 with a scenario-bootstrap 95% interval $[0.897410, 0.982827]$. At the unchanged 0.5 threshold the false-positive rate is 0 and the false-negative rate is 0.85. There are 3 true positives, 17 false negatives, 180 true negatives and no false positives. Good ordering of failures relative to nonfailures therefore coexists with poor failure detection at this particular threshold. It would be inappropriate to describe the classifier as conservative certification of the full financing model.

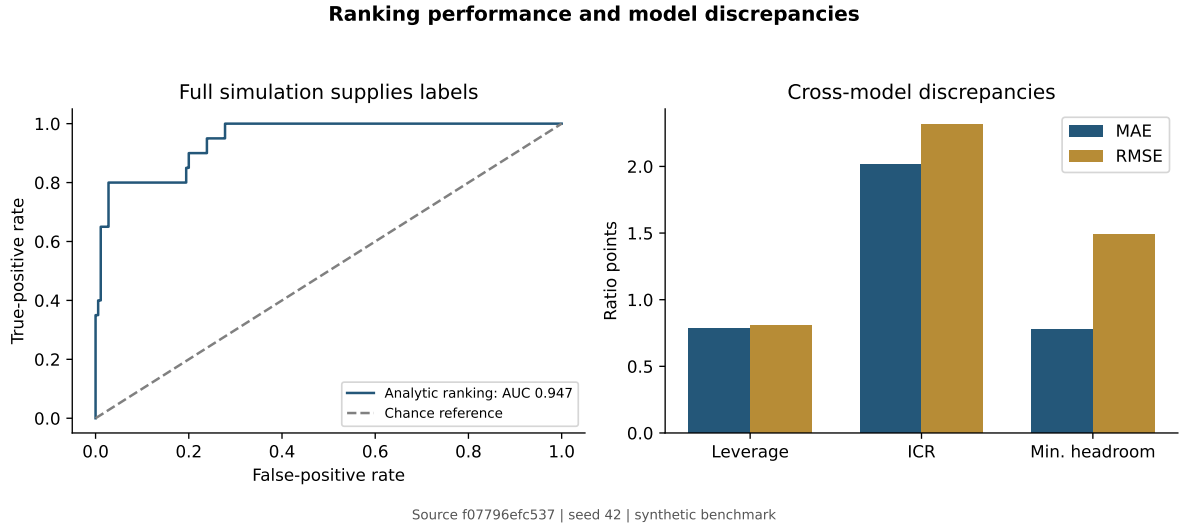


Figure 2: Current method comparison: the analytic score’s ROC against full-simulation financial-failure labels, and cross-model MAE/RMSE. The diagonal is a chance reference, not a reproduced traditional or Bayesian comparator. Error aggregation follows Table 3; headroom and individual ratio errors measure different quantities. Source: benchmark JSON, seed 42.

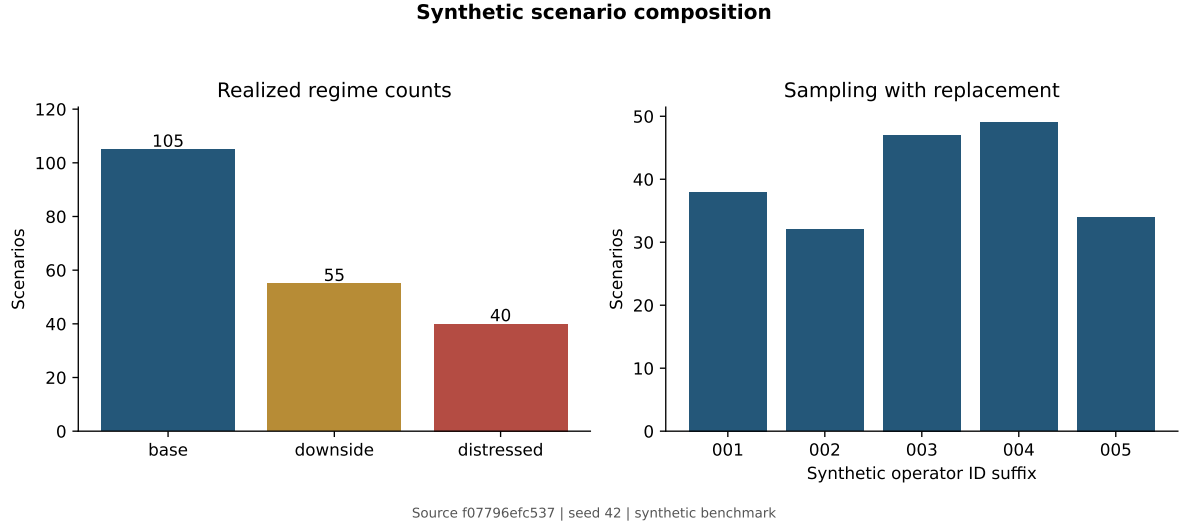


Figure 3: Synthetic benchmark overview: realized regime counts and operator-template sampling frequencies among 200 scenarios. These are synthetic archetypes, not a geographic borrower dataset. Source: benchmark scenario records, seed 42.

7.3 Sensitivity and certification scope

The benchmark fixes annual testing and the two stated covenant thresholds. It does not evaluate quarterly testing, equity cures, hedging interventions, interest-rate floors or optimized covenant packages. The historical IRR sensitivity table has no reproducible counterpart and is not reported as a current result. A future sensitivity experiment must define interventions, matched financing inputs and a sponsor-return protocol before computing effects. Likewise, heuristic diagnostic envelopes in the JSON are not established component budgets for Proposition 1. No bound utilization, false non-certification, or time-point certification rate is inferred from them.

7.4 Breach composition analysis

Twenty of 200 scenarios financially fail; 180 do not. The mutually exclusive priority labels identify 17 payment defaults and 3 other combined reserve/covenant failures. A payment-default label may also involve reserve and covenant failures. The separate flags in Figure 4 preserve that overlap. The JSON field `failed_scenario_count` is zero: it counts execution exceptions, not these 20 financial failures.

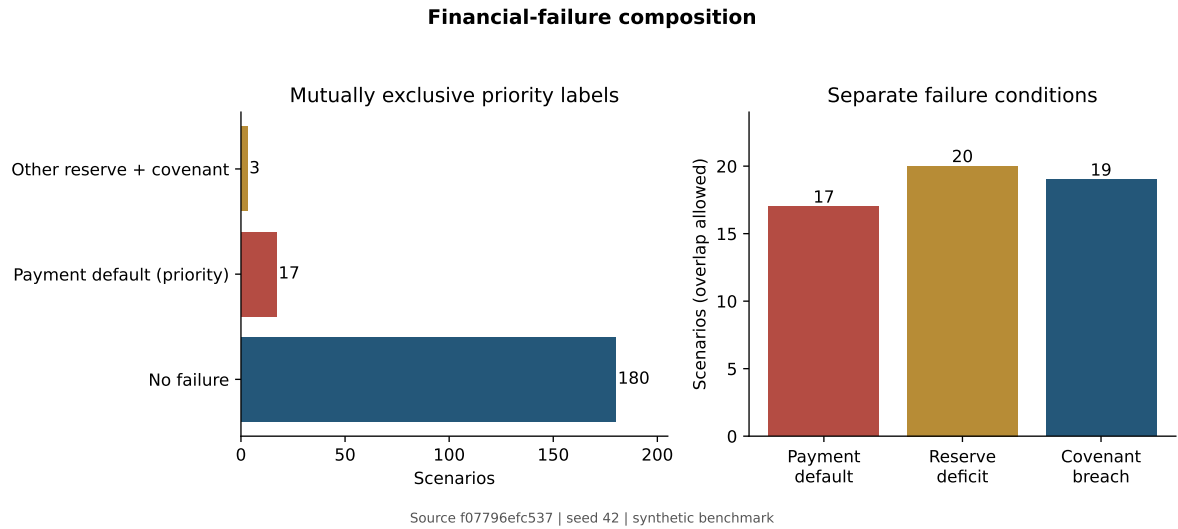


Figure 4: Financial-failure composition. Left: disjoint priority labels, with payment default preceding combined reserve/covenant failure. Right: overlapping payment, reserve and covenant flags; these counts must not be added as if disjoint. Source: full benchmark JSON, seed 42. No base-versus-optimized intervention is claimed.

7.5 Failure-mode diagnostics

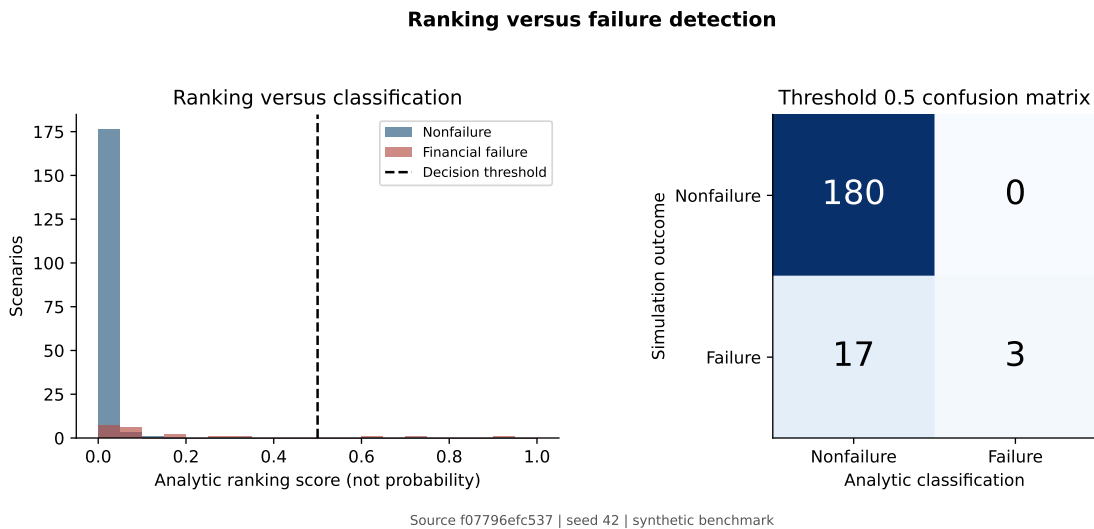


Figure 5: Analytic ranking-score distributions by realized financial failure and the confusion matrix at 0.5. Actual simulation outcomes index rows; analytic classifications index columns. The 17 missed failures make the distinction between ranking discrimination and threshold performance visible. Source: benchmark scenario records, seed 42.

The evidence describes failure detection conditional on the current DGP. It does not identify a causal failure mechanism from growth volatility or validate old percentages assigned to assumption-edge violations. Both model structure and starting inputs can explain discrepancies. Attribution requires controlled one-change-at-a-time or matched-model experiments beyond this evaluation.

8 Accor SA Dual-Convention Analysis

8.1 Illustrative real-company inputs

We retain an illustrative real-company accounting-convention case study using the repository’s reported/reconstructed Accor inputs for 2019–2023. This is not predictive validation. The CSV includes report, page and reconstruction notes; these annotations are provenance claims in the repository, not an independent audit of the cited filings. We reproduce the supplied values without asserting that each value is an audited consolidated-statement line item. The old 2018–2022 table is not the input to the current script.

Table 4: Stipulated Accor case-study inputs (EUR millions) from `data/case_study/accor.csv`. Rent and interest are supplied inputs.

Year	Revenue	EBITDA	Net debt	Leases	Rent	Interest
2019	4049	825	2500	3000	120	190
2020	1621	-34	3350	3200	128	180
2021	2204	364	3100	3050	125	175
2022	4224	675	2800	2900	118	170
2023	5056	1003	2550	2750	110	168

The case-study script allocates supplied net debt 80/20 to two financial-debt fields, sets cash to zero because the supplied quantity is already net, and sets the financial rate to interest expense divided by net debt. The split therefore does not change aggregate financial interest. It assumes a lease rate of 4% and uses supplied lease expense as rent. Including leases on top of the supplied net debt is a stipulated bridge; it requires that the supplied net-debt field exclude those leases for an economic interpretation without double counting. The script does not independently establish that reporting convention from filings.

8.2 Dual-convention ratios and hypothetical thresholds

Table 5: Regenerated Accor ratios. F denotes the stipulated frozen-GAAP helper; 16 denotes the lease-inclusive helper. Undefined 2020 ratios are not zeros.

Year	ICR F	ICR 16	Leverage F	Leverage 16
2019	4.974	2.661	3.030	6.667
2020	–	–	–	–
2021	2.794	1.226	8.516	16.896
2022	4.665	2.360	4.148	8.444
2023	6.625	3.608	2.542	5.284

Across the four positive-EBITDA years, the arithmetic mean ICR changes from 4.764x in the stipulated frozen-GAAP calculation to 2.464x in the IFRS-16 calculation. Mean leverage changes from 4.559x to 9.323x. The negative-EBITDA 2020 observation is excluded from these ratio means and explicitly classified as a failure in both views. These quantities describe the supplied inputs and stipulated conventions; they are not estimates of a predictive model’s accuracy.

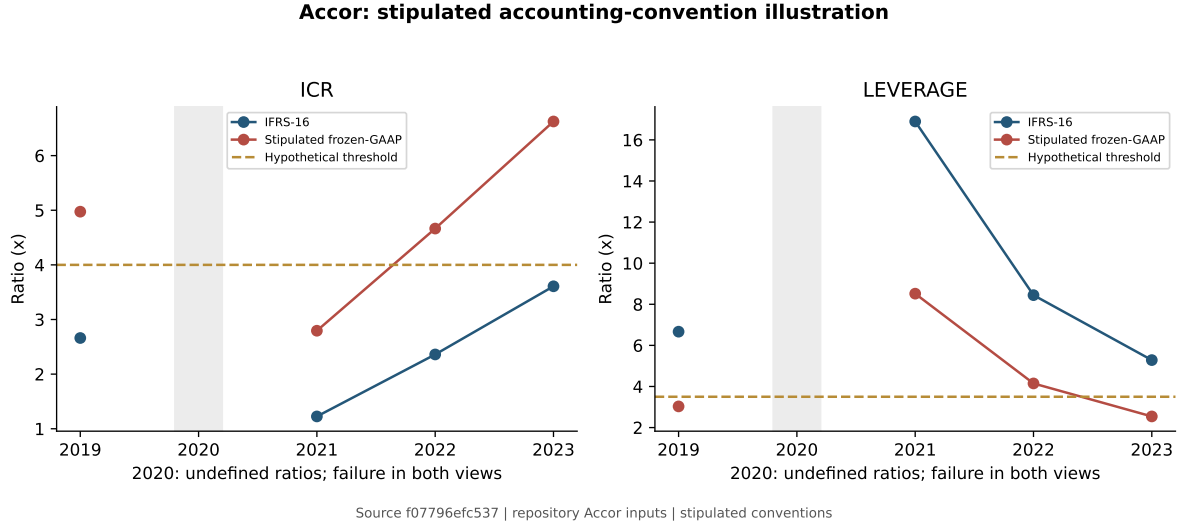


Figure 6: Regenerated Accor accounting-convention illustration. Dashed thresholds (ICR at least 4.0x and leverage at most 3.5x) are hypothetical, not Accor’s actual contractual covenants. The shaded 2020 year has negative EBITDA and undefined ratios; gaps are intentional. Source: repository case-study CSV and the current case-study helper, with its stipulated 4% lease rate.

Including the 2020 failure, the hypothetical tests fail in 5/5 years under the lease-inclusive calculation and 3/5 under the stipulated frozen-GAAP view. The example demonstrates sensitivity to definitions. It does not establish actual contractual breaches, prove that one convention protects a borrower from distress, or show that a redesigned covenant would improve investment returns.

9 Limitations and Future Work

The benchmark is synthetic, uses five archetypes and contains no audited transaction dataset. Its AUC and interval are conditional on the specified DGP, seed, outcome definition and model assumptions. There is no calibrated probability-of-default interpretation or independent borrower test set. The bootstrap samples scenarios rather than operator clusters and does not quantify uncertainty about selecting the templates or the financial model.

Structural, timing and input differences between the two model families preclude interpreting observed discrepancies as pure analytic approximation error. The conditional mathematical guarantees require positive denominator floors and component budgets that the benchmark does not establish. Cash, payment default and reserve adequacy require explicit treatment beyond ratio certification.

The full model has simplified taxes and lease schedules, no complete loss carryforward, no separate right-of-use depreciation schedule, no default recovery model and no arrears catch-up. Negative cash and negative exit residuals are diagnostic; the latter are not limited-liability equity payoffs. Comprehensive input-domain validation is not implemented. The Accor exercise is illustrative rather than predictive validation, and reconstructed inputs require independent filing reconciliation before making company-specific economic claims.

Future work should first match the model inputs, dates, rates and cash definitions, then isolate approximation components. A separate empirical study could evaluate operator-held-out or transaction-held-out performance and score calibration. Return optimization would require an explicit financing decision, default/recovery treatment and a reproducible return protocol. These extensions are research tasks, not findings of this revision.

10 Conclusion

The repaired engine supports a transparent synthetic evaluation of analytic covenant screening against explicit financing dynamics. Ranking discrimination and threshold performance give materially different assessments of this workflow. Conditional mathematical screening and the Accor convention example remain useful when their assumptions and input provenance are explicit. The present evidence does not demonstrate calibrated borrower default probabilities, universal simulation error bounds or an equity-return uplift.

Acknowledgments

The implementation and reproducibility artifacts use the open-source scientific Python and LaTeX ecosystems.

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A Mathematical Proofs

A.1 Exact ratio perturbation

For nonzero $y, \hat{y}$,

$$\frac{x}{y} - \frac{\hat{x}}{\hat{y}} = \frac{x - \hat{x}}{y} + \frac{\hat{x}(\hat{y} - y)}{y\hat{y}}. \quad (25)$$

If both denominator magnitudes are at least $b > 0$, taking absolute values gives $|x/y - \hat{x}/\hat{y}| \leq |x - \hat{x}|/b + |\hat{x}| |\hat{y} - y|/b^2$. A floor on only the unperturbed denominator does not suffice. For example, $x = \hat{x} = 1$, $y = 1$, $\hat{y} = 1/2$ gives difference 1, exceeding the erroneous first-order bound $|\hat{y} - y|/y^2 = 1/2$. Thus the historical inequality cannot be treated as a finite-perturbation theorem.

Proof of Proposition 1. Apply (25) with $(x, y) = (E, I)$ and then (N, E) . Use the respective common denominator floors i and e and the stated analytic numerator bounds. The triangle inequality gives $|N^a - N| \leq |D^a - D| + |L^a - L| + |C^a - C|$ for matched definitions. With nonnegative identical rates and timing, $|I^a - I| \leq r_d |D^a - D| + r_L |L^a - L|$. Substitution yields the result. $\square$

Proof of Proposition 2. Every x satisfying $\text{Lev}_t^a(x) + \epsilon_{\text{lev}}(t) \leq c_1^{\text{lev}}$ also satisfies the inequality with c_2^{lev} . All other constraints and the objective are unchanged, so set inclusion and monotonicity of the supremum follow. No continuity or strict improvement is implied. $\square$

Proof of Proposition 3. At each date, $\text{ICR}_t \geq \text{ICR}_t^a - \epsilon_{\text{icr}}(t) > c^{\text{icr}}$ and $\text{Lev}_t \leq \text{Lev}_t^a + \epsilon_{\text{lev}}(t) < c^{\text{lev}}$. These are the only events certified by the premise. $\square$

B Theoretical Assumptions

Assumption A1 (Matched definitions). The two theoretical paths have the same testing dates, accounting definitions, units, recurrence coefficients and interest timing. Any differing cash flows, initial values or inputs have explicitly established error budgets.

Assumption A2 (Positive denominator floors). Both EBITDA paths exceed a known $e > 0$ and both applicable interest paths exceed a known $i > 0$ at each test. If either floor fails, no ratio certification is made.

Assumption A3 (Finite perturbations). Component perturbations are bounded in absolute value; analytic numerator upper bounds are finite. Cash error enters net debt. Debt, lease and rent conventions are fixed for each ratio comparison.

Assumption A4 (Reduced-form recurrences). For the recurrence lemma below, both paths use the same debt coefficient $a = 1 + r_d \geq 0$, sweep $s \geq 0$, and the same nonnegative projection. Lease paths use a common $q \in [0, 1]$ and explicitly bounded addition differences.

These assumptions replace incompatible historical growth, CPI and quarterly testing assertions. They do not claim that the full benchmark meets the matched-model conditions.

C Explicit Error Bound Derivations

Lemma 1 (Matched recurrence stability). *Consider the matched recurrences*

$$D_{t+1} = \max(0, aD_t - sF_t), \quad \widehat{D}_{t+1} = \max(0, a\widehat{D}_t - s\widehat{F}_t),$$

with $|F_t - \widehat{F}_t| \leq \delta F_t$. Then

$$|D_t - \widehat{D}_t| \leq a^t \epsilon_D(0) + s \sum_{k=0}^{t-1} a^{t-1-k} \delta F_k. \quad (26)$$

For lease recurrences $L_{t+1} = \max(0, (1-q)L_t + A_t^L)$ with identical q ,

$$|L_t - \widehat{L}_t| \leq (1-q)^t \epsilon_L(0) + \sum_{k=0}^{t-1} (1-q)^{t-1-k} \delta A_k^L. \quad (27)$$

Proof. The map $x \mapsto \max(0, x)$ is 1-Lipschitz. Hence the debt difference at $t+1$ is at most a times the difference at t plus $s\delta F_t$. Iteration proves (26); replacing a with $1-q$ and the forcing error with δA_t^L proves (27). $\square$

These formulas propagate known forcing errors; they do not establish those errors from an arbitrary FCF conversion assumption. If a forcing bound has the form $\delta F_k \leq b(1+g)^k$, its sum is

$$\sum_{k=0}^{t-1} a^{t-1-k} (1+g)^k = \begin{cases} \frac{a^t - (1+g)^t}{a - (1+g)}, & a \neq 1+g, \\ ta^{t-1}, & a = 1+g. \end{cases} \quad (28)$$

For a separately stipulated CPI-indexed payment model, preserve the conventional algebraic schedule $L_{t+1} = (1+r_L)L_t - P_t$, with $P_t = P_0(1+\pi)^t$. It is not the current run-off implementation. For $-1 < \pi_{\min} \leq \pi, \widehat{\pi} \leq \pi_{\max}$, the mean-value theorem yields

$$|P_t(\pi) - P_t(\widehat{\pi})| \leq P_0 t (1 + \pi_{\max})^{t-1} |\pi - \widehat{\pi}| \quad (t \geq 1), \quad (29)$$

and payment differences can be propagated as in (26) with coefficient $1 + r_L$. This avoids the historical expression with denominator $1 - (1 + \pi_{\min})^{-T}$, which is negative for $\pi_{\min} = -0.02$ and $T > 0$. No numeric CPI bound is transferred to the full simulation.

Finally, for constant growth $g, \widehat{g} \in [g_-, g_+]$ with $g_- > -1$ and the same starting EBITDA,

$$|E_0(1+g)^t - E_0(1+\widehat{g})^t| \leq |E_0| t (1+g_+)^{t-1} |g - \widehat{g}|. \quad (30)$$

This finite-perturbation bound follows from the mean-value theorem. A capex conversion parameter changes FCF, not this EBITDA growth identity. None of these statements supplies a numerical certification rate for the empirical experiment.

D Benchmark Generator Details

The archive contains the complete input assumptions for both model families, scenario-level outcomes and the 1,000 annual simulation and analytic observations. The figure generator reconstructs the seed-42 draws and checks the analytic scores, simulated maximum leverage and minimum coverage against the archived benchmark before plotting. Figures and tables use the current numerical output; the original figures' filenames are retained to make replacement traceable.

E Computational Environment and Reproducibility

Source revision (manuscript and generator): f07796efc537468122804e6eae25b1a4f284efbb.

Benchmark revision: f07796efc537468122804e6eae25b1a4f284efbb.

Python 3.14.0; seed 42; 200 scenarios; five annual observations per scenario; 500 scenario-bootstrap attempts.

Environment SHA-256: 8adf35826a18275799f7a7fe3e5a9723631ddf53f7b7159965aad9dcd974c0eb. The source revision identifies the exact committed manuscript and generation code. A subsequent artifact-only commit stores the generated provenance, figures, benchmark and PDF: a commit cannot embed its own final hash in its own contents. The artifact manifest records actual SHA-256 checksums, including the environment export. No release tag, DOI deposit, historical environment hash or fixed timing claim is asserted. The repository license is MIT.

From the repository root, with dependencies installed, run:

```
python -m ruff check .
python -m ruff format --check .
python -m pytest -q --cov=src/lbo --cov-fail-under=52
python -m analysis.run_benchmark --seed 42
python -m analysis.scripts.generate_paper_v2_figures --run-benchmark
latexmk -pdf -outdir=paper paper/ifrs16_lbo_ssrn_v2.tex
```

The figure command archives a fresh full report in `results/paper_v2/benchmark_seed42.json`; the standalone benchmark otherwise overwrites its output directory. When `latexmk` is absent, use repeated `pdflatex` passes or the documented Tectonic fallback in `paper/REPRODUCE_V2.md`. Financial quantities are reproducible for fixed code, inputs, seed and dependencies; wall-clock measurements are environment-dependent. Local validation and PDF inspection records are archived separately from the scientific claims.